



Slime-resistant marine anti-biofouling coating with PVP-based copolymer in PDMS matrix

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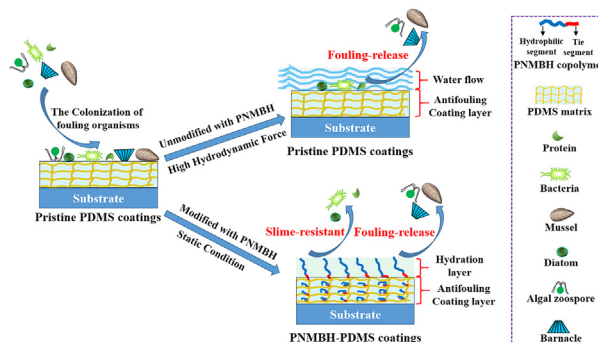
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HIGHLIGHTS

- New copolymer (PNMBH) can improve the anti-biofouling property of PDMS.
- Easy and straightforward incorporation of PNMBH with PDMS matrix.
- Maintaining anti-biofouling performance for ≥ 4 months in marine field.
- Possessing both slime-resistant performance and fouling-release performance.

GRAPHICAL ABSTRACT



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ABSTRACT

Poly(dimethylsiloxane) (PDMS) has been widely used as the fouling-release coatings due to its low surface energy. However, for marine anti-biofouling applications, current PDMS-based materials with hydrophobic property still suffer from marine slime attachment. Therefore, the design of novel amphiphilic coatings by modifying PDMS with hydrophilic polymers has been widely explored. In this work, based on poly(N-vinylpyrrolidone) (PVP), a novel hydrophilicity-active poly(N-vinyl-2-pyrrolidone-co-methyl methacrylate-co-butyl acrylate-co-2-hydroxyethyl methacrylate) (PNMBH) was synthesized using a robust free radical polymerization method. Next, amphiphilic coatings were prepared by crosslinking different amounts of PNMBH with PDMS at room temperature. The resultant PNMBH-PDMS coatings were characterized with water contact angle tests, presenting their increased hydrophilicity and the optimal PNMBH content (10%). Moreover, the anti-biofouling performance of these coatings was evaluated using proteins (bovine serum albumin and fibrinogen), bacteria (Gram-negative *Escherichia coli* and Gram-positive *Staphylococcus aureus*), and marine diatom (*Navicula parva*). Compared with pristine PDMS coating, PNMBH-PDMS coatings could reduce $\sim 82\%$ adhesion of fibrinogen, $\sim 97\%$ adhesion of *Staphylococcus aureus*, and $\sim 97\%$ adhesion of diatom, the major composition of marine slime. More importantly, PNMBH-PDMS coatings were also tested in the marine field, and results showed their excellent anti-biofouling performance for at least 4 months in the ocean. This work developed a novel amphiphilic

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coating with appealing marine anti-biofouling performance and might provide new opportunities for marine coating applications.

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1. Introduction

Marine biofouling, the undesired settlement and accumulation of marine organisms on the submerged surfaces, has become a global problem impeding the development of maritime transportation and aquatic industries (Callow and Callow, 2011). In the shipping industry, biofouling can cause various detrimental impacts, including the increased roughness of ship-hull surfaces, the loss of vessels maneuverability, further inducing the increased maintenance cost, fuel consumption, and greenhouse gas emission (Chambers et al., 2006). In addition, during the shipping process, foreign creatures may be introduced into the non-native maritime spaces, leading to ecological unbalance (Occhipinti-Ambrogi, 2007; Occhipinti-Ambrogi and Savini, 2003). Currently, the utilization of anti-fouling coatings on the submerged surfaces is a state-of-the-art strategy to prevent marine biofouling. Since mid-1960s, biocide-based anti-biofouling coatings have been widely used. For example, the marine coatings using tributyltin (TBT) as biocide showed an excellent performance to resist fouling attachment for as long as 10 years (Omae, 2003). However, due to the adverse effects of TBT on the nontargeted organisms in marine environment, it has been absolutely forbidden in 2008 (Lejars et al., 2012; Magin et al., 2010). Afterwards, coatings based on copper biocide have taken the marine anti-biofouling market (Thomas and Brooks, 2010). Unfortunately, these coatings still possess potentially harmful effects to marine environment (Dickenson et al., 2017). It has been reported that the copper concentrations exceeding 3.1 µg/L are highly toxic toward nontargeted aquatic organisms such as teleosts, sea urchins, and crustaceans (Ciriminna et al., 2015; Mochida et al., 2006).

As an environmentally-friendly alternative, fouling-release (FR) coatings based on poly(dimethylsiloxane) (PDMS) are commercially available (Martinelli et al., 2012). Due to the low surface energy and high elastic modulus of PDMS materials, many kinds of macro-organisms (such as oyster and barnacle) exhibit difficulty to attach firmly on their surfaces, leading to easy detachment under hydrodynamic forces (Yang et al., 2014; Yasani et al., 2014). However, PDMS-based coatings are unable to resist the accumulations and growth of marine slime layers, which are mainly composed of bacteria, diatoms, and a matrix of secreted extracellular polymeric substances (EPS) (Hawkins et al., 2014). Because these biofouling can attach more strongly to the hydrophobic PDMS surfaces via hydrophobic interactions, and thus they are difficult to detach even by large hydrodynamic shear force at a high navigational speed (Zhou et al., 2014).

To develop FR coatings with slime-resistant performance, hydrophilic materials have been introduced into PDMS (Galli and Martinelli, 2017; Jiang et al., 2017). For example, poly(ethylene glycol) and zwitterionic polymers are both common hydrophilic materials, which can tightly bind water molecules to form a hydration layer on their surfaces (Cao and Jiang, 2012; Wang et al., 2018; Zhang et al., 2006). This hydration layer can serve as a barrier to strongly resist protein adsorption and bacterial adhesion, thus inhibit the initial process of the fouling organism settlement (Banerjee et al., 2011; Cao et al., 2016, 2013a, 2013b; Jiang and Cao, 2010; Kingshott et al., 2002). Meanwhile, they have been used to modify PDMS for amphiphilic coatings with anti-biofouling performance including slime resistance (Keefe et al., 2012; Stafslin et al., 2015). However, the current hydrophilic materials still suffer

from problems such as limited stability (Roosjen et al., 2004; Yang et al., 2017) or complex preparation procedure (Bodkhe et al., 2015), hampering their widespread applications in marine anti-biofouling coatings.

Poly(*N*-vinylpyrrolidone) (PVP) is a water-soluble and highly biocompatible polymer with stable physicochemical properties (Gallardo et al., 2017; Haaf et al., 1985; Sulek et al., 2012). Moreover, the surfaces modified with PVP possessed excellent antifouling properties that can resist protein adsorption and cell adhesion (Liu et al., 2010; Ran et al., 2011; Telford et al., 2010; Wu et al., 2009, 2012). PVP possesses hydrophilic cyclic amide units in the side chains and hydrophobic methylene/methane moieties in the backbone (Laukkanen et al., 2004), resulting in their superior solubility in both aqueous and organic solvents. Therefore, the monomer can efficiently react with hydrophilic, hydrophobic monomers or polymer segments.

In this study, we developed a facile method to prepare a novel crosslinked network coating, which was based on modified PVP copolymer (PNMBH) in PDMS matrix. This novel coating could efficiently resist slime attachment, while maintaining its FR performance. Firstly, PNMBH was synthesized by free radical polymerization, using the monomer *N*-vinyl-2-pyrrolidone (NVP), methyl methacrylate (MMA), butyl acrylate (BA), and 2-hydroxyethyl methacrylate (HEMA) (Scheme 1a). Secondly, PNMBH was directly crosslinked with PDMS matrix at room temperature to form the amphiphilic PNMBH-PDMS coatings (Scheme 1b). Lastly, the anti-biofouling performances of the novel coatings were evaluated in the laboratory and marine environments. This work developed the environmentally-friendly coatings with highly effective anti-biofouling performance, offering a new solution for the current problems induced by marine biofouling.

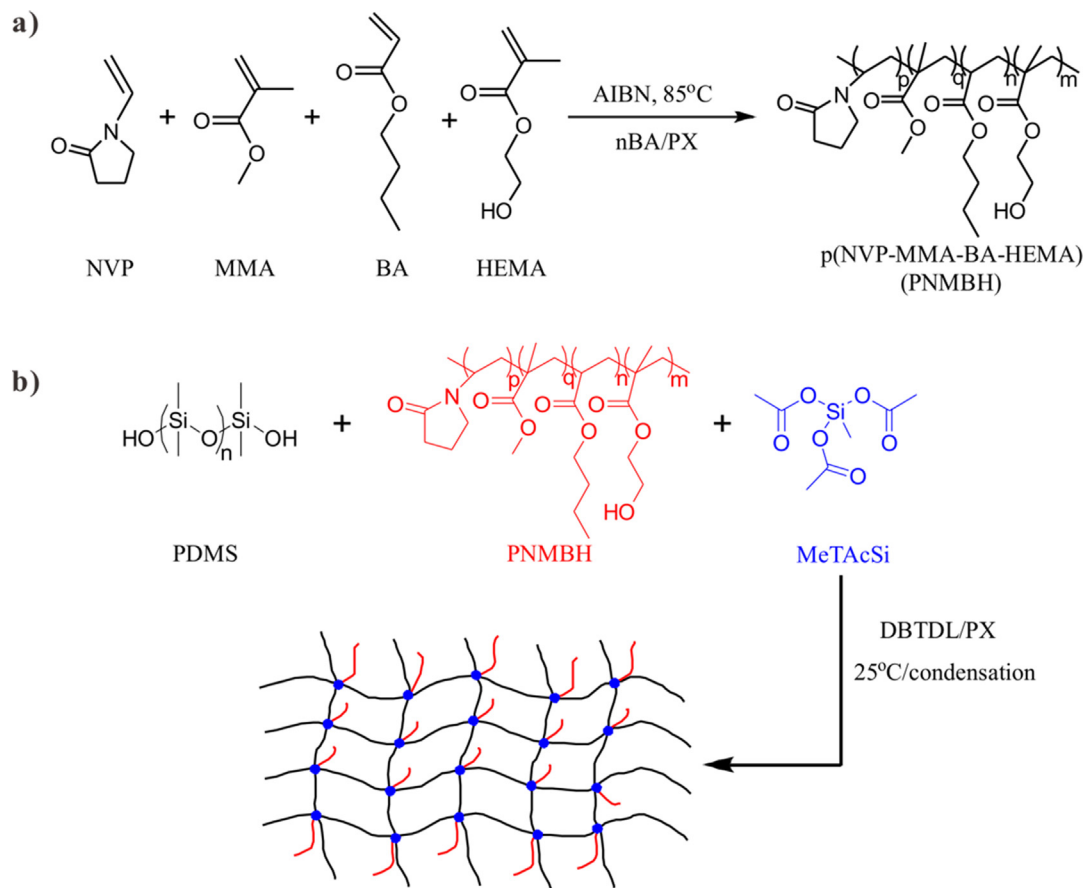
2. Materials and methods

2.1. Materials

Methyl methacrylate (MMA), butyl acrylate (BA), 2-hydroxyethyl methacrylate (HEMA) and *N*-vinyl-2-pyrrolidone (NVP) were all purchased from Acros Organics. Methyltriacetoxysilane (MeTAcSi) and dibutyltin dilaurate (DBTDL) were both obtained from Adamas. Silanol-terminated polydimethylsiloxane (HO-PDMS-OH, $M_w = 26,000$ g/mol) was purchased from TCI Shanghai. *p*-xylene (PX), *n*-butyl acetate (nBAC), dichloromethane, and butyl alcohol were all purchased from Concord Technology Tianjin. The polyvinyl chloride (PVC) panel was purchased from Cangzhou Sansu Co., Ltd.

2.2. Synthesis of Poly(*N*-vinyl-2-pyrrolidone-co-methyl methacrylate-co-butyl acrylate-co-2-hydroxyethyl methacrylate) (PNMBH)

A mixture of dimethylbenzene (6 mL) and *n*-butyl acetate (6 mL) was stirred at 85 °C in a round bottom flask. Then, BA (20 mmol, 2.563 g), MMA (10 mmol, 1.0012 g), NVP (60 mmol, 6.6684 g), HEMA (10 mmol, 1.3014 g), and AIBN (1.2 mmol, 0.2 g) were all dissolved in dimethylbenzene (6 mL) and *n*-butyl acetate (6 mL) and they were dropwise added to the flask. After reaction for 8 h at 85 °C, the product was precipitated with diethyl ether and dried in the vacuum at 60 °C, achieving a yield of 87%.



Scheme 1. (a) Scheme for the synthesis of PVP-based copolymer (PNMBH) (NVP: 60%, MMA: 15%, BA: 15%, HEMA: 10% in molar ratio). (b) The formation of the amphiphilic coatings using PNMBH crosslinked with PDMS matrix at room temperature.

2.3. Coating preparation

Different contents of PNMBH (wt%) were mixed with HO-PDMS-OH (2 g) in a mixture of dimethylbenzene (2 mL) and n-butyl acetate (2 mL). Then, the mixtures were stirred constantly for 30 min at room temperature to become homogeneous. Next, 150 μ L MeTAcSi and 20 μ L DBTDL were both added into the mixtures, and stirred for 10 min at room temperature to obtain the original coating materials. Glass slides were ultrasonically cleaned using acetone and ethanol for 10 min, respectively, and then rinsed thoroughly with deionized water. The original coating materials were cast on the clean glass slides and then placed into an enclosed drying cabinet with ventilation overnight. Finally, the PNMBH-PDMS coatings were cured at room temperature for 24 h.

2.4. Characterization

The chemical structure of PNMBH copolymers were determined by ^1H NMR spectroscopy. The spectra were recorded at room temperature on a VARIAN INOVA spectrometer (500 MHz) as a solution in CDCl_3 , and chemical shifts were reported in parts per million with tetramethylsilane as an internal standard. The chemical structures of PNMBH-PDMS coatings after soaking in water for 1 h were determined by Fourier transform infrared spectrometer (FTIR, Bruker, Germany) under 25 °C. The number-average molecular weight (M_n) and polydispersity index (M_w/M_n) of PNMBH copolymers were determined by gel permeation chromatography (GPC, waters 1515/2414, Waters Company, USA). THF was used as the eluting solvent with a flow rate of 1 mL/min, and polystyrene was used

as the standard for calibration. The water contact angles (WCA) were measured using an optical contact angle measuring device (Zhongyikexin Technology Co. Ltd JC2000DM, China) under ambient environment. The probe liquids were the distilled water (7 μ L). The elapsed time between drop placing and measurement was 10 s. The determined contact angle of each sample was the average value of six different positions.

2.5. Protein adsorption

2.5.1. Bovine serum albumin labelled with fluorescein isothiocyanate (BSA-FITC) adsorption

BSA-FITC solution (50 μ L) was diluted in 10 mL of PBS (the final concentration was 0.05 mg/mL). Samples were spun cast on silicon wafers. Then, these silicon wafers were incubated in BSA-FITC solutions in a dark room for 2 h and then rinsed with PBS thoroughly (Bi et al., 2006). Then, samples were observed using an inverted fluorescence microscope (Nikon Eclipse Ti-S; Japan).

2.5.2. Enzyme linked immunosorbent assay (ELISA) measurement

Enzyme-linked immunosorbent assay (ELISA) test was used to measure fibrinogen (Fg) adsorption according to previous literature (Yang et al., 2016). All samples were cut to 5 mm diameter disks and placed into a 48-well plate. Then, 0.5 mL of 1 mg/mL fibrinogen from human plasma was added to the wells and incubated at 37 °C for 90 min. Each sample was rinsed nine times with 1 mL PBS and incubated in BSA (1 mg/mL in PBS) solution at 37 °C for 90 min to block the areas unoccupied by Fg. The samples were rinsed nine times with PBS, transferred to new wells, and

incubated in 0.5 mL PBS solution containing 5.5 $\mu\text{g/mL}$ horseradish peroxidase (HRP) conjugated anti-fibrinogen at 37 °C for 30 min. The samples were rinsed nine times with 1 mL PBS and transferred into clean wells, followed by the addition of 0.5 mL of 0.1 mol/L citrate-phosphate buffer (pH 5) containing 1 mg/mL chromogen o-phenylenediamine (OPD) and 0.03% hydrogen peroxide. After incubation at 37 °C for 20 min, the enzyme-induced color reaction was quenched by adding 0.5 mL of 1 mol/L H_2SO_4 in each well, and the absorbance at 490 nm was determined by a microplate reader. The absorbance of pristine PDMS samples was equivalent to 100% for calculating relative adsorption values.

2.6. Bacterial adhesion

Two representative strains, *Escherichia coli* (*E. coli*, gram-negative) and *Staphylococcus aureus* (*S. aureus*, gram-positive) were used for the bacterial adhesion tests on coatings according to previous literature (Zhu et al., 2017). Disks with 1 cm diameter of different samples were incubated in LB broth containing 1×10^6 cfu/mL of microbial suspensions in 24-well plates. After incubation at 37 °C for 6 h, these samples were washed three times with 1 mL of PBS, and 100 μL of the last washing fluid was spread on the LB agar plates to incubate for 24 h. After incubation, the numbers of colonies were counted for each plate.

Green fluorescent protein-expressing *Escherichia coli* (GFP-*E. coli*) was also used. GFP-*E. coli* was cultured overnight at 37 °C in Luria-Bertani (LB) media with 50 $\mu\text{g/mL}$ kanamycin sulfate. The cells were harvested by centrifugation at 4 °C and 2000 rpm for 15 min, and washed three times in PBS buffer (pH 7.4). The solution was adjusted to an optical density (OD) at 600 nm of 1.0. The different disks were immersed in bacterial solution ($1\text{--}1.5 \times 10^9$ cfu/mL) and incubated at 37 °C and 150 rpm for 2 h. After incubation the disks were washed with PBS three times during 15 min in order to remove the non-adhered bacteria. Bacteria adhesion to the different surfaces was imaged with inverted fluorescence microscope (Nikon Eclipse Ti-S) excited by blue light.

2.7. Marine diatom adhesion

Marine diatom adhesion test was conducted according to previous literature (Zhang et al., 2013). Diatom *Navicula parva* was collected from Taipingjiao beach of Qingdao. Diatom suspension was prepared as described and its concentration was adjusted to 1×10^5 cells mL^{-1} by filtered and sterilized seawater. All lab assay slides were placed in a vessel containing 1 L cell suspension. The vessel was placed in climatic cabinet at 20 °C, in which the luminance was controlled at 7000 Lx. After incubated for 24 h, all slides were gently washed in sterilized artificial seawater (prepared by Mocledon prescription) to remove unattached cells, and then immediately subjected to observe by optical microscope. Diatom cells were counted in 30 fields of view from each of three replicates to provide the statistical average number of adhered diatoms with 95% confidence interval.

2.8. Pseudobarnacle adhesion strength measurements

The measurement of barnacle adhesion strength in shear was based on American Society for Testing and Material (ASTM) D5618-94 (1994). This method used a hand-held force gauge to mimic barnacle pillars from coatings. The mimetic barnacle pillars ($\phi = 20$ mm, $h = 10.5$ mm) were stuck to the surface of coating samples by epoxy glue (Aibida 6005, epoxy resin). Force was applied to the base of the barnacle pillars, parallel to the surfaces, at a rate of approximately 4.5 N s^{-1} until the mimetic barnacle was detached (Guo et al., 2019).

2.9. Marine field tests

The field tests were performed from May 2018 to September 2018 at the Qingdao Port (36°0839'N, 120°3118'E) in China near Yellow Sea. The seawater has an average temperature of 25.5 °C with a maximum of 29 °C and a minimum of 18 °C, during summer and autumn. The salinity is about 31‰, and the pH ranges from 7.6 to 8.2. The samples were coated onto the carbon steel reinforced epoxy resin panels ($300 \times 200 \times 2 \text{ mm}^3$) and immersed into seawater at a depth of 0.2–2.0 m. After a certain period of time, the coated panels were taken out of the sea, carefully washed with seawater and photographed. The anti-biofouling activity was evaluated according to American Society for Testing and Materials (ASTM) D6990-05 (2011).

3. Results and discussion

3.1. Coating preparation and characterization

The synthetic route of PNMBH was presented in Scheme 1a. In the PNMBH polymer, PVP was the hydrophilic segment, while PMMA and PBA were both hydrophobic segments that could facilitate the integration of PNMBH with the PDMS matrix. Meanwhile, PHEMA segment possessed reactive hydroxyl groups which could chemically crosslink with PDMS. The chemical structure of PNMBH was demonstrated by ^1H NMR spectrum, as shown in Fig. S1. The peak at 3.61 ppm (peak d) was corresponding to the $-\text{OCH}_3$ from MMA. The protons of the $-\text{OCH}_2-$ were in the low field 4.03 and 3.78 ppm (peak b, c and d) in BA and HEMA. The protons at 3.20 ppm (peak c), 2.32 ppm (peak a) and 1.59 ppm (peak b) were assigned to the $\text{N}-\text{CH}_2-$, $-\text{CO}-\text{CH}_2-$ and $\text{CH}_2-\text{CH}_2-\text{CH}_2$ in NVP. Furthermore, using quantitative analysis by ^1H NMR spectrum, composition ratio of PNMBH copolymer was 0.574: 0.157: 0.172: 0.097 (PVP: PMMA: PBA: PHEMA in molar ratio). The molecular mass of PNMBH was determined by GPC, revealing that its average molecular weight was 24,045 g/mol and the polydispersity index (PDI) was 2.19.

The crosslinking reaction route to form amphiphilic PNMBH-PDMS coatings was shown in Scheme 1b. Four ratios of PNMBH/PDMS were prepared and tested by ATR-FTIR (Fig. S2). The characteristic absorption peak that located at 1020 cm^{-1} represented the Si-O groups of PDMS. Notably, the surface chemical structure of coatings was analyzed by ATR-FTIR, and the peaks around 1180 cm^{-1} corresponded to the stretching vibration of $-\text{COO}-$, a segment of PNMBH. This result indicated that PNMBH had been successfully crosslinked in the PDMS matrix and migrated to the coating surfaces.

The water contact angle (WCA) tests were used in this work to evaluate the hydrophilicity of these coatings (Fig. 1). Compared with the pristine PDMS coating (WCA, $117.7 \pm 1.0^\circ$), WCAs of the PNMBH-PDMS coatings significantly decreased (all below 70°), indicating the increased hydrophilicity after modification of PNMBH. The increased hydrophilicity was caused by the migration of hydrophilic PVP segment to coating surfaces. It should be noticed that the coating with 10% of PNMBH possessed the lowest WCA ($43.8 \pm 1.1^\circ$), suggesting the optimal ratio of PNMBH/PDMS. The results revealed that the novel PVP-based copolymer, PNMBH, could dramatically improve the hydrophilicity of PDMS materials, highly useful to marine antifouling coatings for slime resistance.

3.2. Protein adsorption

After exposed to seawater, the submerged substrate starts to get contaminated by a biomolecular film mainly containing proteins (Banerjee et al., 2011; Iguerb et al., 2008; Lejars et al.,

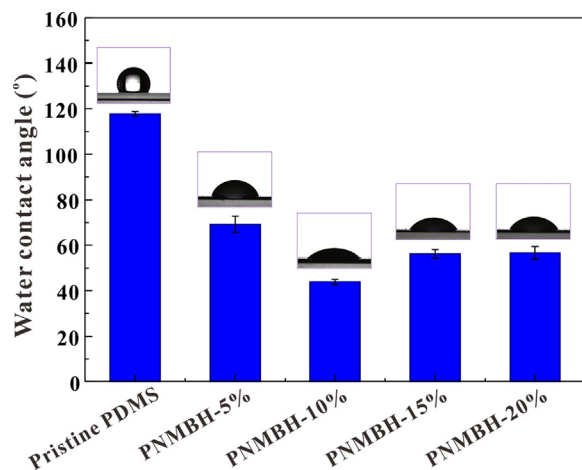


Fig. 1. Water contact angle tests of different coatings.

2012). Therefore, it is the primary and critical issue to resist protein absorption for the development of anti-biofouling coatings. In this work, two representative types of proteins (BSA-FITC and fibrinogen) were chosen to qualitatively and quantitatively evaluate the protein-resistance performance of the coatings. Fig. 2 showed that few fluorescence signals could be observed on all the PNMBH-PDMS coating surfaces, while a large amount of BSA-FITCs were found on the surface of pristine PDMS coating. Moreover, fibrinogen adsorption on these coatings was tested using ELISA measurement, and the results were presented in Fig. 3. Compared with the pristine PDMS coating (100%), fibrinogen adsorption on the PNMBH-PDMS coating with 10% PNMBH could be significantly reduced by 82%. These results demonstrated that the amphiphilic PNMBH-PDMS coatings could efficiently resist the protein adsorption.

3.3. Bacterial adhesion

In marine environment, attached bacteria in biofilm can supply abundant nutrient for further settlement of fouling macro-organisms (Lejars et al., 2012). Therefore, two representative bacteria strains, gram-negative bacterium (*E. coli*) and gram-positive bacterium (*S. aureus*), were chosen in this work to evaluate the bacterial adhesion resistance of these coatings (Fig. 4). Compared with

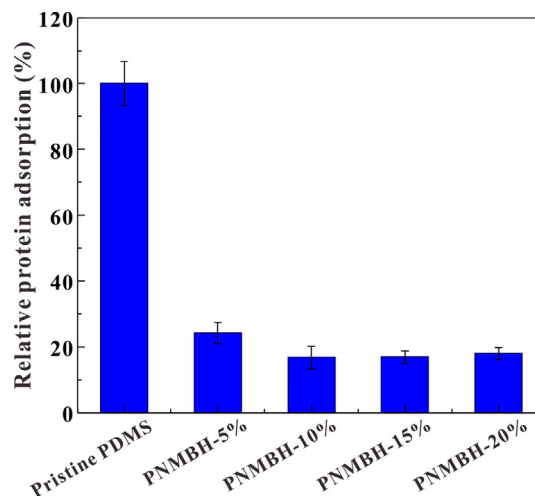


Fig. 3. Fibrinogen adsorption tests using ELISA on different coatings. All samples were incubated in Fibrinogen solution for 1.5 h.

a large number of bacterial colonies on PDMS, much fewer bacterial colonies could be observed on PNMBH-PDMS coating surfaces. The best performance (~91% fewer *E. coli* and ~97% fewer *S. aureus*) was achieved with 10% PNMBH coating, consistent with the protein adsorption tests. The antibacterial results were further confirmed by the adhesion test of green fluorescent protein-expressing *E. coli*, as shown in Fig. 5. These results demonstrated the strong bacterial resistance ability of the PNMBH-PDMS coatings.

3.4. Diatom adhesion

PDMS materials are hydrophobic and thus possess poor abilities to inhibit the attachment of marine slime (Holland et al., 2004). Diatom is one representative type of slimes and it has a hard 'SiO₂' cell wall (frustule) (Landoulsi et al., 2011). Via hydrophobic interaction, diatom can easily attach on the PDMS surfaces which also contain a large amount of Si. Therefore, we used diatom *Navicula parva* to evaluate their adhesion on PNMBH-PDMS coatings (Fig. 6). It could be found that these amphiphilic coatings were all able to efficiently resist diatom adhesion compared with pristine PDMS coating, demonstrating that they could overcome the

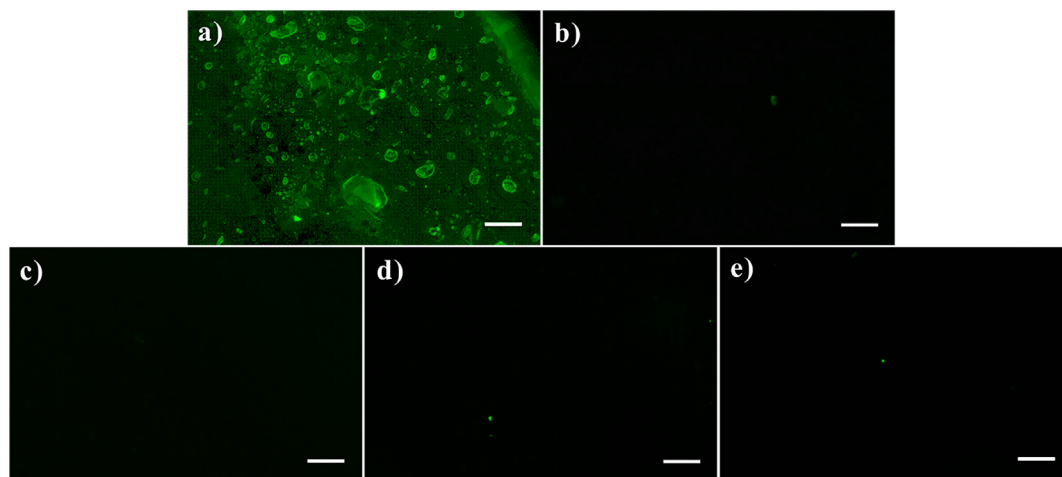


Fig. 2. Fluorescent images of BSA-FITC adsorption on (a) pristine PDMS coating, (b) PDMS coating with 5% PNMBH, (c) PDMS coating with 10% PNMBH, (d) PDMS coating with 15% PNMBH, and (e) PDMS coating with 20% PNMBH. All samples were incubated in BSA-FITC solution for 2 h. Scale bar: 100 μ m.

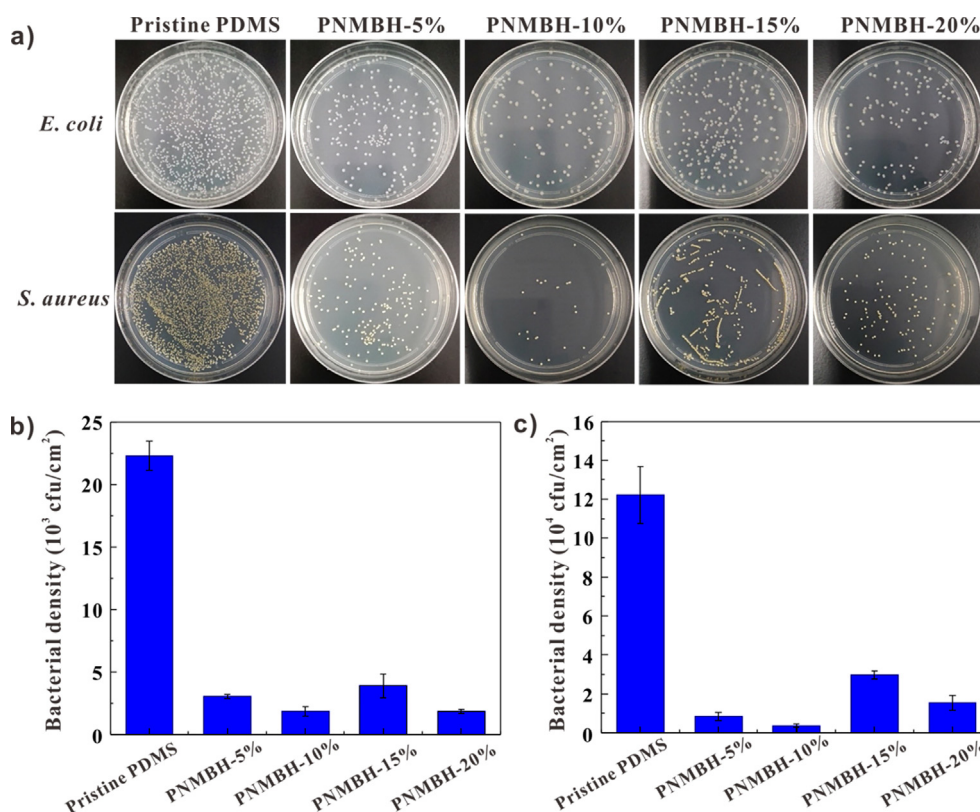


Fig. 4. Adhesion of two representative strains (*E. coli* and *S. aureus*). (a) Photo images of bacterial colonial morphologies on the surfaces of pristine PDMS coating and PNMBH-PDMS coatings. (b) The density of the attached bacterial numbers of *E. coli*. (c) The density of the attached bacterial numbers of *S. aureus*. All the samples were incubated in bacterial suspensions with 1×10^6 CFU/mL at 37 °C for 6 h.

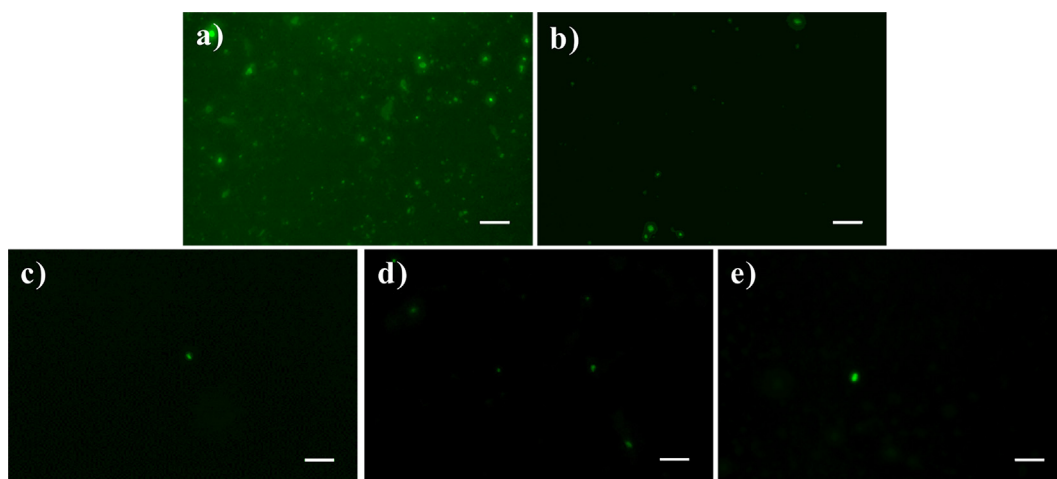


Fig. 5. Fluorescence microscopy images of green fluorescent protein-expressing *Escherichia coli* (GFP-*E. coli*) attached on the surfaces of (a) pristine PDMS coating, (b) PDMS coating with 5% PNMBH, (c) PDMS coating with 10% PNMBH, (d) PDMS coating with 15% PNMBH, (e) PDMS coating with 20% PNMBH. All samples were immersed in bacterial solution ($1-1.5 \times 10^9$ cfu/mL) and incubated at 37 °C for 2 h. Scale bar: 100 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

disadvantage of PDMS materials in marine anti-fouling applications.

3.5. Pseudobarnacle adhesion strength measurements

Due to the FR property of the PDMS coatings, barnacle attached on their surfaces can be easily released with hydrodynamic forces. In this work, in order to evaluate the FR performance of the PNMBH-PDMS coatings, the pseudobarnacle adhesion strength

measurements were performed using ASTM standard test method (ASTM D5618-94, 1994). This measurement can provide a good reference of the FR performance of the coatings. Fig. 7a presented the mimetic barnacle shear strength on these coatings. Maximum shear strength and adhesion strength in shear for different samples were showed in Fig. 7b. Compared with pristine PDMS coating, the mimetic barnacle adhesion strengths of all PNMBH-PDMS coatings observably declined. Notably, the barnacle adhesion strength on the surfaces of 10% PNMBH coating was 0.0082 MPa, 67% lower

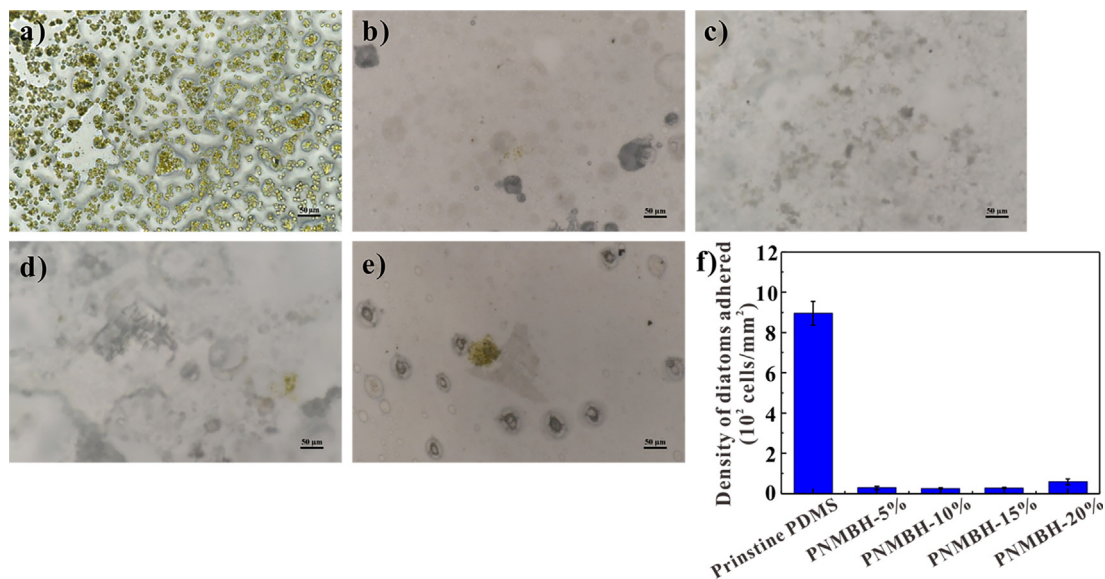


Fig. 6. Adhesion of diatom *Navicula parva*. The photo images of diatom colonial morphologies on the surfaces of (a) pristine PDMS coating, (b) PDMS coating with 5% PNMBH, (c) PDMS coating with 10% PNMBH, (d) PDMS coating with 15% PNMBH, (e) PDMS coating with 20% PNMBH after 24 h incubation. (f) The density of diatom adhesion. Scale bar: 50 μ m.

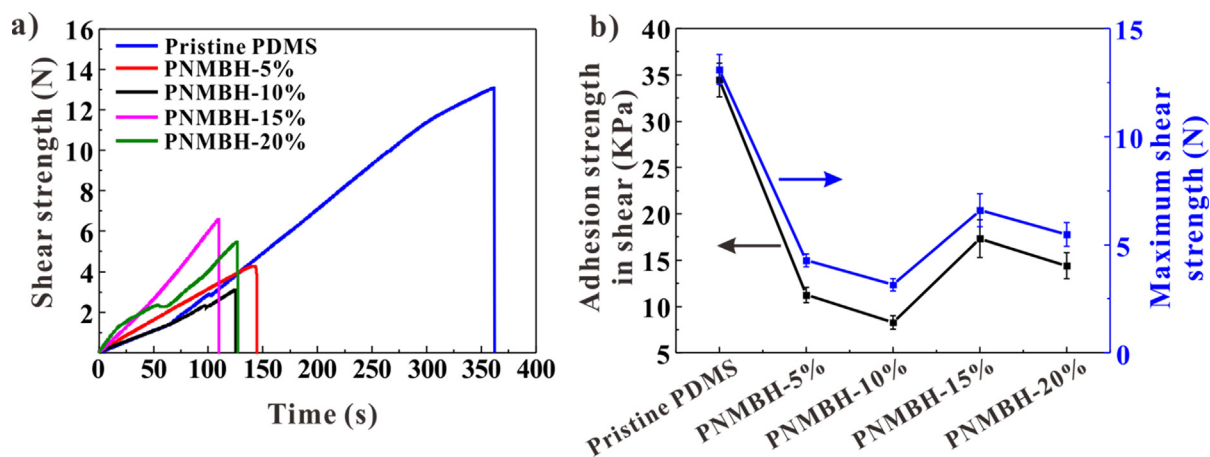


Fig. 7. The pseudobarnacle adhesion strength measurements on different coatings. (a) The shear strength on the coatings versus the time; (b) maximum shear strength and barnacle adhesion strength in shear for different coatings.

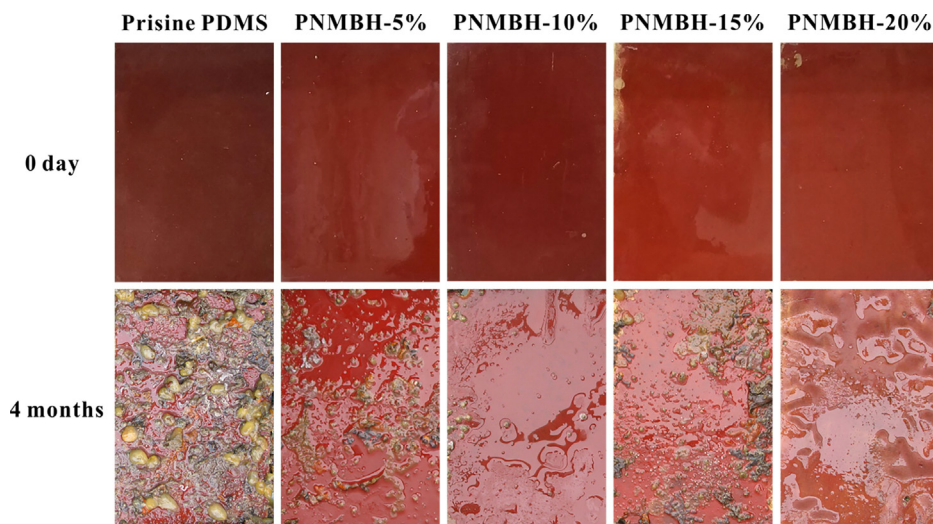


Fig. 8. Images of blank control (PVC plate), panels coated with pristine PDMS coating and PNMBH-PDMS coatings before and after immersion in seawater for 4 months (May–September 2018, Qingdao, China).

Double cooperative defense mechanism

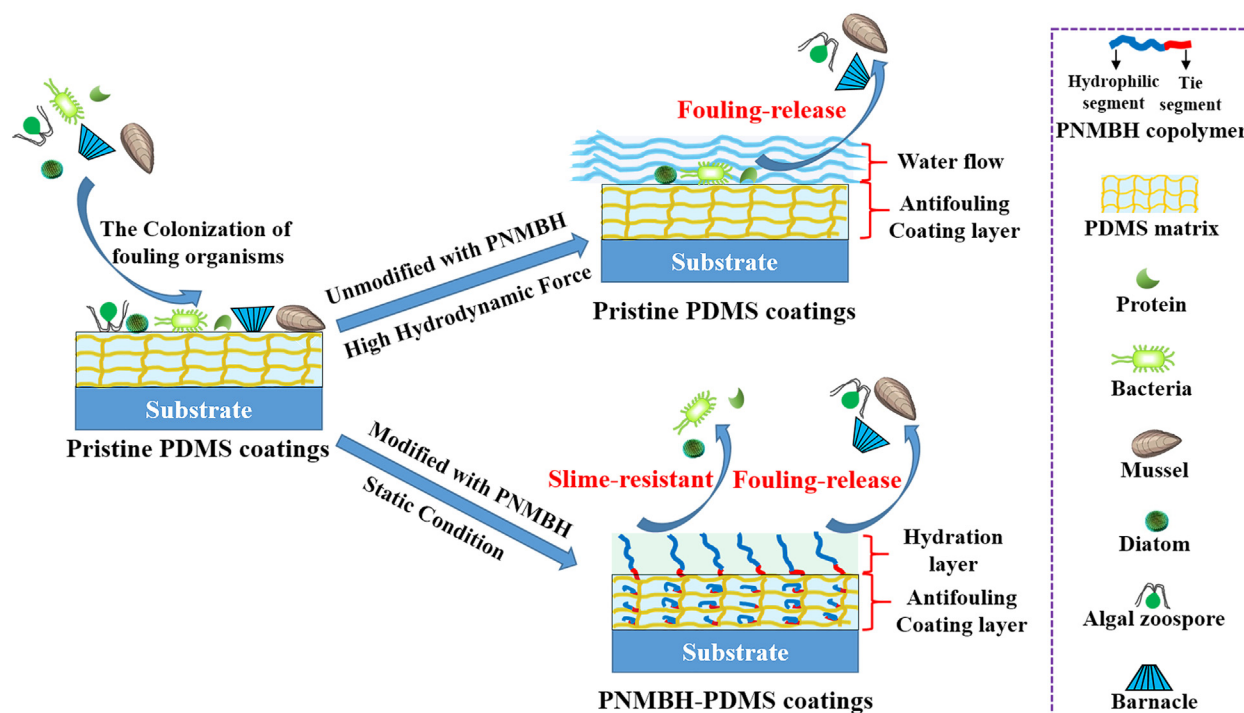


Fig. 9. Cooperative defense of PNMBH-PDMS coating against biofouling.

than that of pristine PDMS coating, indicating that the modification of PNMBH could significantly enhance the FR performance of PDMS to resist barnacle adhesion.

3.6. Marine field tests

Marine environment contains numerous and various biofouling components. Therefore, at present, the absolute resistance to marine biofouling is still a huge challenge. In this work, we evaluated the anti-biofouling performance of the PNMBH-PDMS coatings in marine field (Yellow Sea, China), as shown in Fig. 8. After 4 months, PNMBH-PDMS coatings showed much better performance compared to the blank control (PVC panel) and the pristine PDMS coating which have been covered with a mass of fouling organisms (including diatom, algae, molluscs and barnacle). This result also demonstrated the high stability of these novel coatings, suggesting their promising potential for marine antifouling applications.

3.7. Potential mechanism

According to the above results, a potential mechanism was proposed for the anti-biofouling performance of the PNMBH-PDMS coatings (Fig. 9):

The amphiphilic PNMBH-PDMS coatings developed in this work can integrate the FR property of PDMS with the slime-resistance property of hydrophilic PVP. On one hand, PDMS materials possess low surface energy, roughness, glass transition, and modulus, and they can detach a considerable amount of biofouling macro-organisms (such as mussel and barnacle) via hydrodynamic forces generated by movement of the ship through the water or by gentle 'grooming' devices. On the other hand, when the PVP-based copolymer coatings are immersed in seawater, the hydrophilic PVP segments can migrate to the coating surfaces, and a hydration layer can form to prevent the slime (containing proteins, bacteria and diatoms) attachment. Therefore, the novel amphiphilic coat-

ings show cooperative defense against biofouling, which means that the FR coatings possess the slime-resistant performance.

4. Conclusions

In summary, we developed a type of novel marine antifouling coatings based on the directly crosslinked networks of PNMBH and PDMS at room temperature. The coatings could not only decrease the barnacle adhesion strength, but also prevent protein adsorption, bacteria adhesion, and diatom adhesion compared to pristine PDMS coatings. Therefore, they possessed cooperative defense against biofouling organisms, namely FR and slime-resistant properties. Notably, these coatings exhibited a favorable anti-biofouling efficacy even in the marine environment. Therefore, this work provided an eco-friendly strategy for the battle against marine biofouling.

Declaration of Competing Interest

The authors declare no competing interests.

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Appendix A. Supplementary material

The Supporting Information including ^1H NMR spectrum of PNMBH in CDCl_3 and FT-IR spectra of coatings are attached in

the Supplementary data. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2019.06.042>.

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